

Comparative analysis of the rainfall products over Elmanagil – Sudan (2001 – 2023) :

CHIRPS, GPM IMERG , and ERA5-LAND: Climatology, Trend, and Bias Correction

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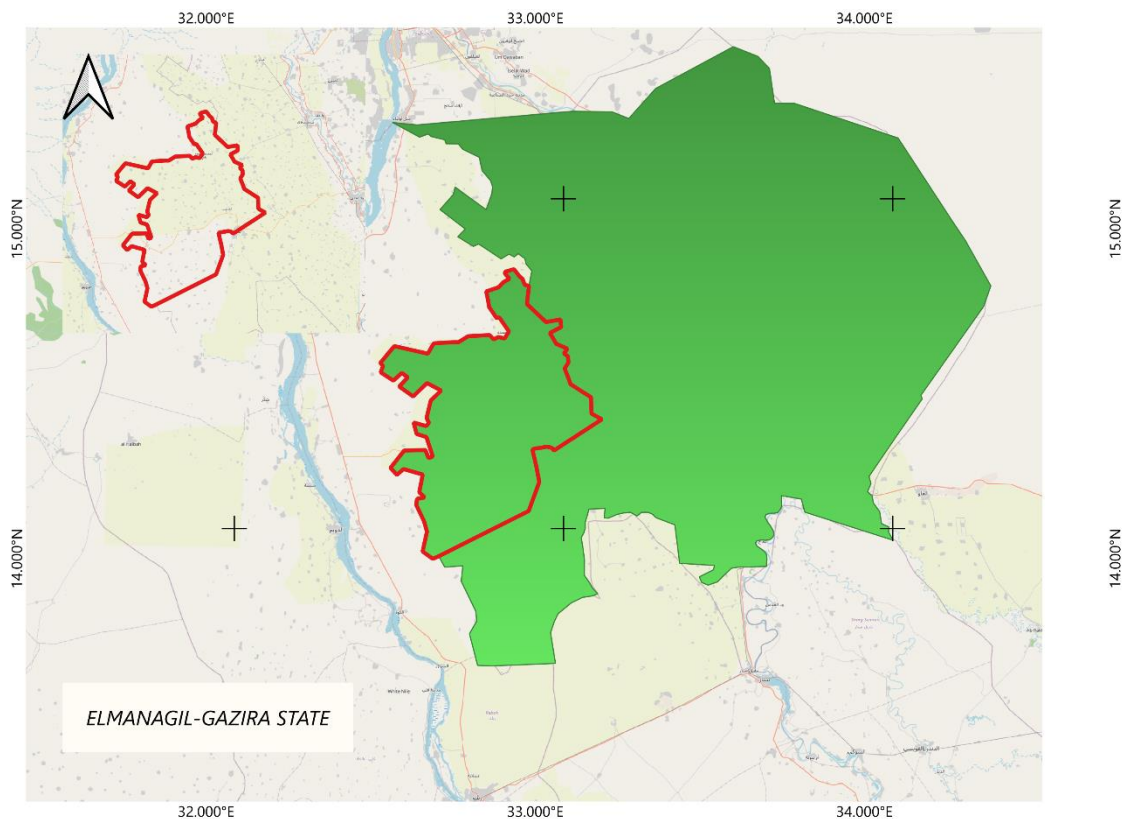
October 2026

Objective:

This project was conducted to evaluate the suitability of three widely used rainfall products – CHIRPS, GPM IMERG, and ERA5-Land for hydroclimatic analysis over Elmanagil locality, Sudan part of the Gezira agricultural scheme in the eastern Nile sub-basin. The analysis covers seasonal rainfall climatology, inter-annual trends, and the effect of bias correction on the weakest-performing product. The work was carried out as preparation for the ENTRO 4th Batch RCRP internship on “climate data and forecasting for hydrological analysis”.

Study area:

Elmanagil locality lies within the Gezira agricultural scheme, Sudan, an area where rainfall availability directly affects supplementary rainfed agriculture and irrigation planning. The study area boundary was uploaded as an asset in Google Earth Engine where the rest of the analysis was performed.



Data and Methodology:

Rain fall products used:

- CHIRPS
- GEM IMERG
- ERA5-Land

Processing steps:

- Monthly basin-mean rainfall series (2001-2023) extracted in Google Earth Engine.
- Quality control: unit verification, plausibility checks, and missing values screening
- Monthly climatology
- Trend analysis using the Mann Kendall test and Sens slope estimator on JJAS annual totals .
- Product comparison against CHIRPS as reference : bias, RMSE, and pearson correlation(r)
- Bias correction of ERA5-Land using monthly empirical quantile mapping, calibrated on 2001-2015 and validated on 2016-2023.

Results:

Monthly rainfall climatology: All three products capture the same single peak rainy season, GPM IMERG and CHIRP shows closely aligned peak month totals, while ERA5-Land consistently underestimates rainfall during the peak of the season.

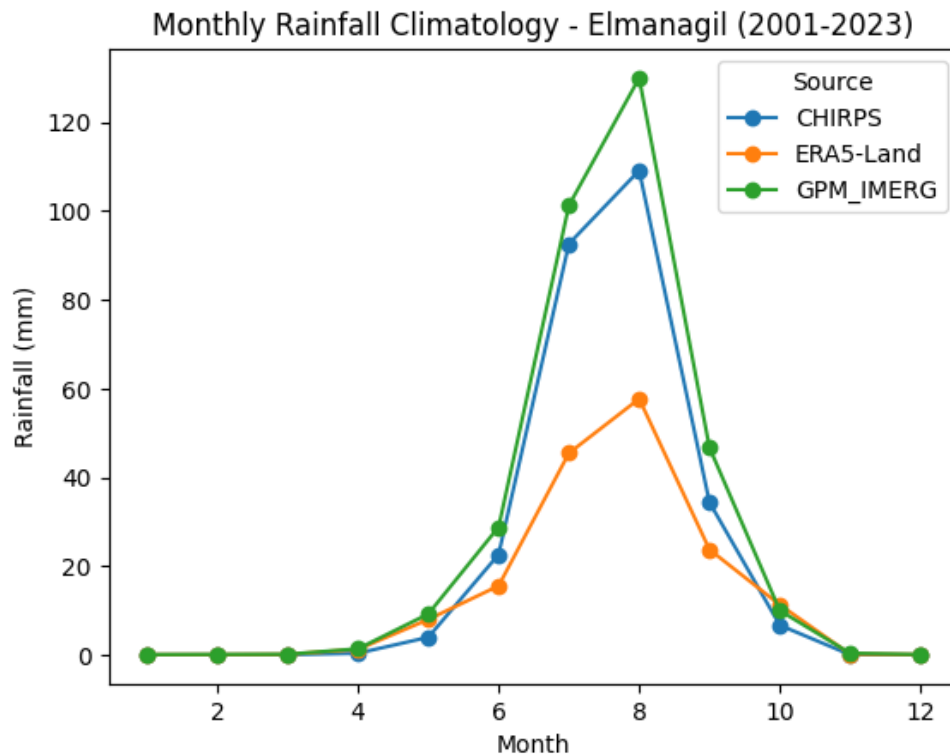


Figure 2. monthly rainfall climatology – CHIRPS, ERA5-Land, GPM IMERG (2001-2023)

Seasonal (JJAS) Totals by year: Year to year JJAS totals vary across the three products, with generally consistent identification of wetter and drier years, though the products do not always agree closely in magnitude.

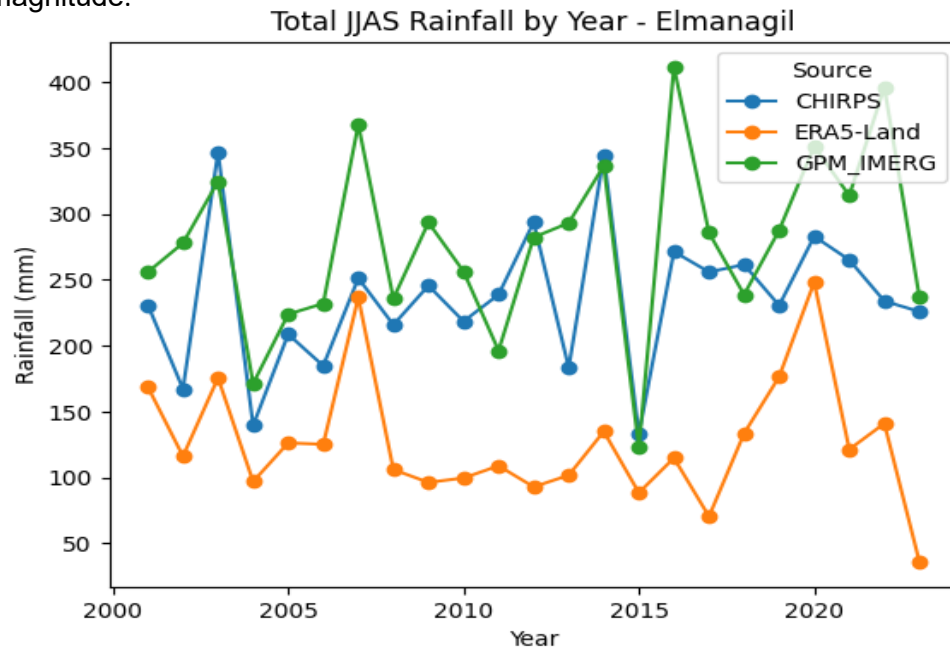


Figure 3. total JJAS rainfall by year, all products

Trend analysis (Mann-Kendall):

No product showed any significant trend in JJAS seasonal rainfall totals over 2001-2023 ($p > 0.05$ for all three) the direction of Sen's slope differs between products, which reflects genuine uncertainty in long term rainfall trend estimation for this area when relying on a single product.

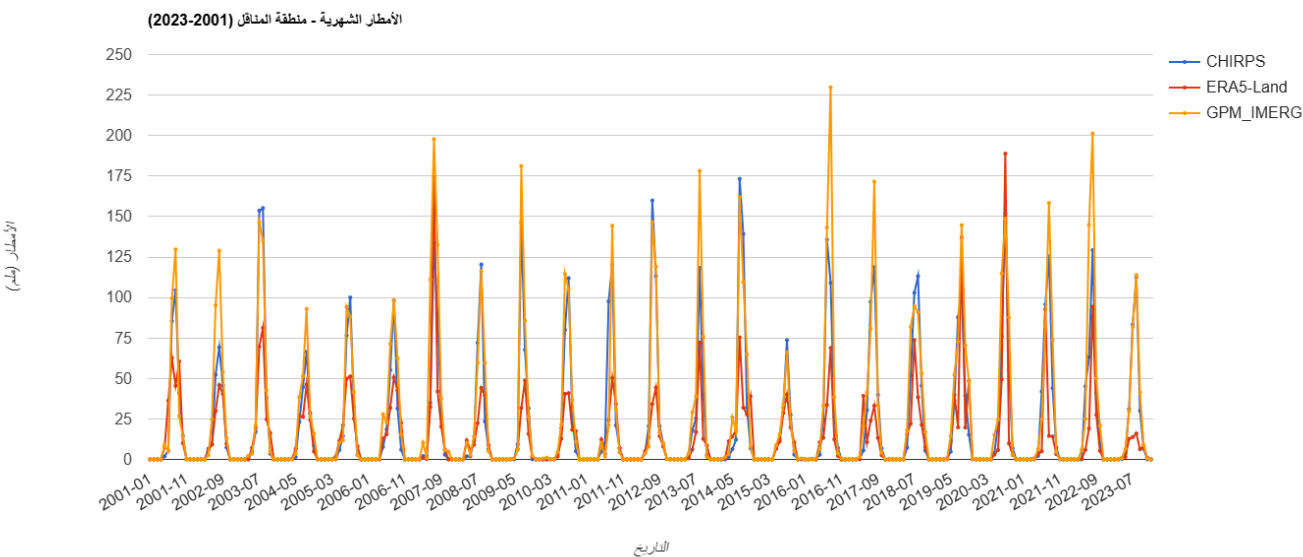
Product	Trend	p-value	Sen's slope (mm/year)
CHIRPS	No trend	0.205	+1.84
ERA5-Land	No trend	0.635	-1.03
GPM IMERG	No trend	0.113	+3.47

Product comparison:

GEM IMERG shows the strongest agreement with CHIRPS, both in correlation and error magnitude. ERA5-Land shows weaker agreement and a clear negative bias.

Product	Bias (mm)	RMSE (mm)	Correlation (r)
ERA5-Land	-8.88	26.88	0.78
GPM IMERG	+4.85	17.47	0.94

Figure 4. Scatter plots – ERA5-Land vs. CHIRPS and GPM IMERG vs. CHIRPS



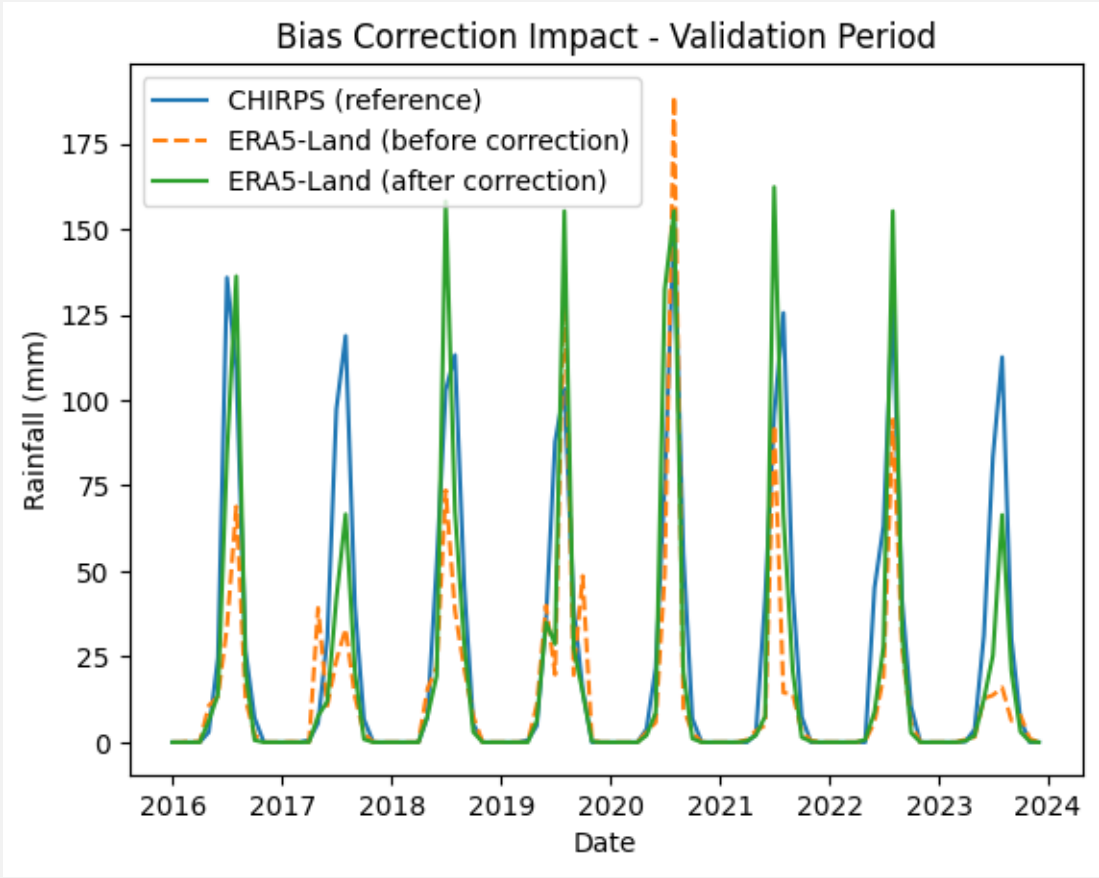
Bias Correction of ERA5-Land

Monthly empirical quantile mapping was applied to ERA5-Land, calibrated on 2001–2015 and validated on the independent 2016–2023 period. The correction reduced RMSE by approximately 24.5%, indicating that a substantial part of the ERA5-Land underestimation is a systematic, correctable bias rather than random error.

Metric	Before Correction	After Correction	Improvement
RMSE (validation period, mm)	29.18	22.02	≈ 24.5%

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[Figure 5. ERA5-Land before and after bias correction vs. CHIRPS – validation period]



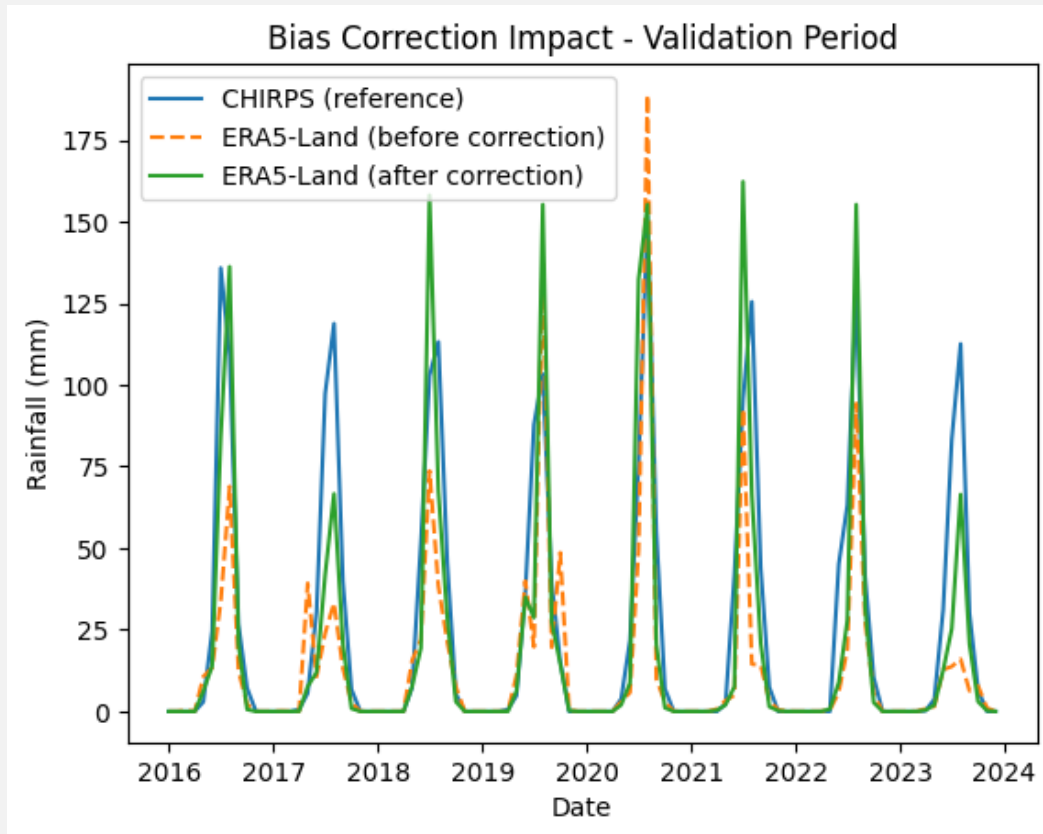
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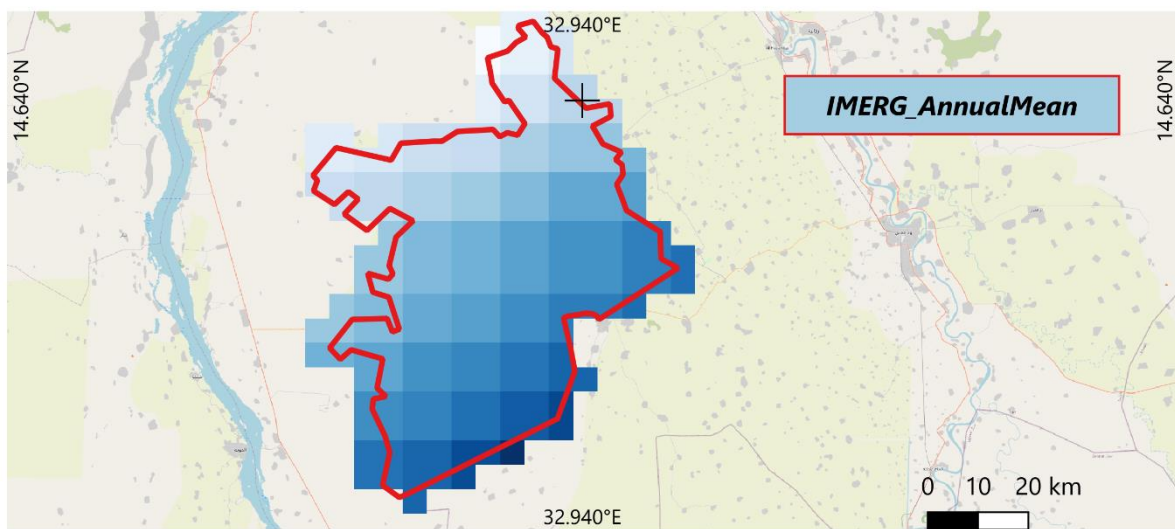
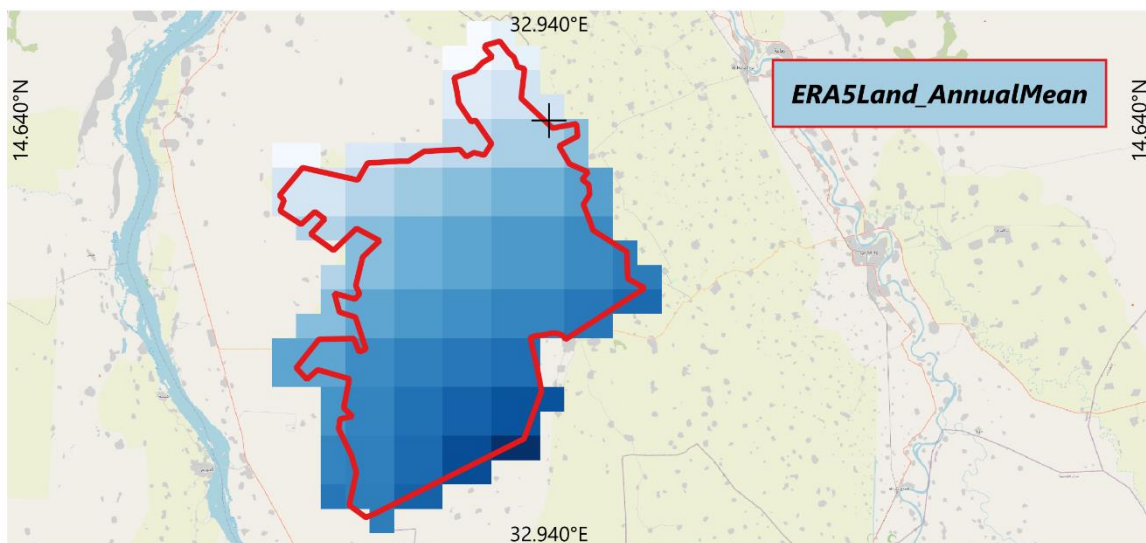
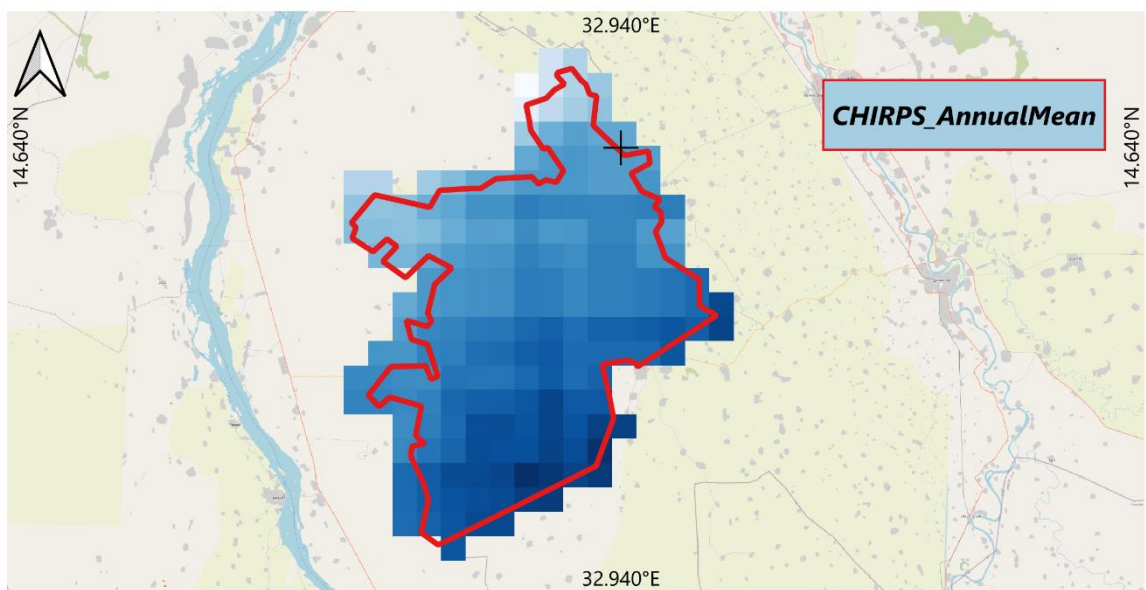
[Figure 5. ERA5-Land before and after bias correction vs. CHIRPS – validation period]



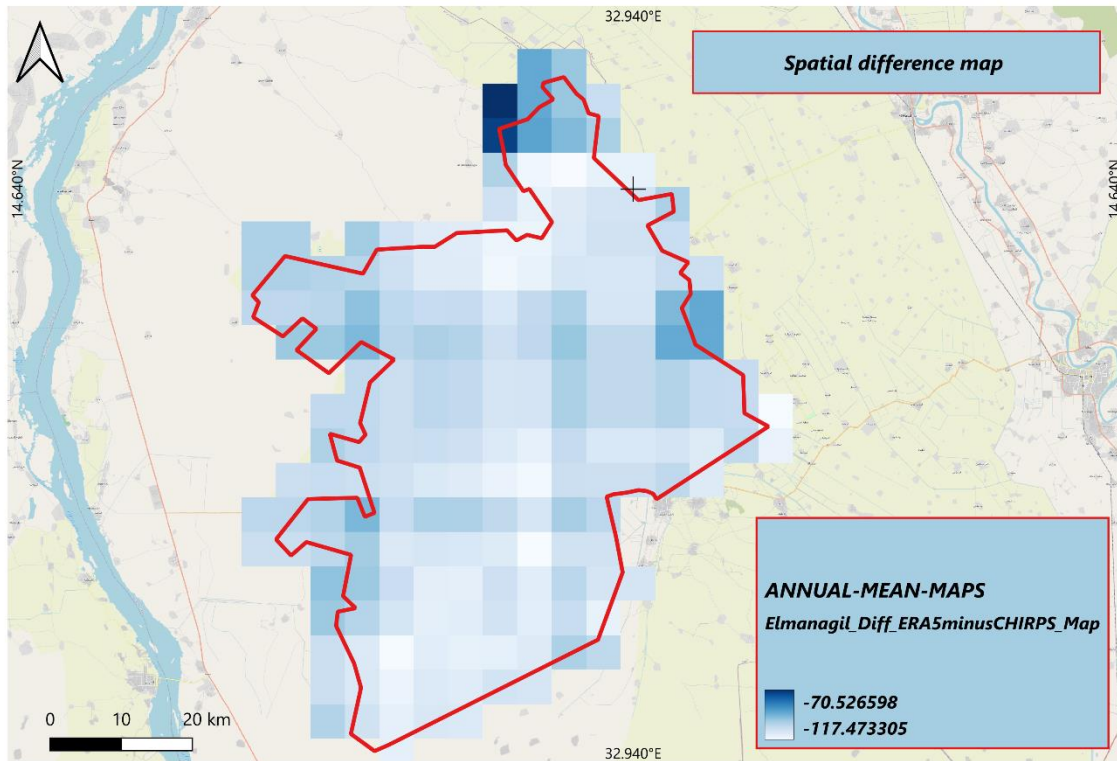
Spatial Distribution:

Annual mean rainfall maps for each product, and a spatial difference map (ERA5-Land minus CHIRPS), illustrate how the bias is distributed across the Elmanagil study area rather than only as a single basin-average figure.

[Figure 6. Annual mean rainfall maps – CHIRPS, GPM IMERG, ERA5-Land]



[Figure 7. Spatial difference map – ERA5-Land minus CHIRPS]



Discussion:

The results indicate that GPM IMERG is the most consistent product with CHIRPS over Elmanagil and is therefore the more reliable choice where satellite rainfall is needed for hydrological applications in this area. ERA5-Land, while valuable for its full suite of land-surface variables (soil moisture, evapotranspiration, temperature) that are not directly available from CHIRPS or IMERG, requires bias correction before its rainfall field is used quantitatively in hydrological modelling for this semi-arid, short-rainy-season environment.

Conclusion and Recommendation:

GPM IMERG is recommended as the primary rainfall input for hydrological analysis over Elmanagil given its strong agreement with CHIRPS ($r = 0.94$, $RMSE = 17.47$ mm). ERA5-Land remains useful for auxiliary variables (evapotranspiration, soil moisture, temperature) but should be bias-corrected — using the monthly quantile mapping approach.

References

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